CURIOUS SPAROOWS · ENCYCLOPEDIC INDEX

TECHNOLOGY

How a sharpened stone became a smartphone

Electronic devices · home appliances · mechanical devices and machines · tools · technology by field: textiles to AI, medicine to space · landmark register · timeline · inventors and institutions · awards · ideas · practical guide

Parts	Sections	Sub-sections	Entries	Register rows
13	39	27	233	110

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How to Use This Booklet

This booklet is a map of technology: electronic devices, home appliances, mechanical devices and machines, tools, and technology in every field from textiles to AI. Every entry tells you, in one or two lines, what the thing is and why it matters.

Every entry has an ID (such as TEC-03.1 for the refrigerator). The same ID is the entry's address on curioussparoows.com, where each one is growing into its own page.

How this booklet is organised

Parts: the intro questions, the master map, devices and appliances, machines and tools, fields (two Parts), registers, people and institutions, ideas, the practical guide, reviews, navigation and sources.

Sections (TEC-01 to TEC-27): the main topics.

Sub-sections and entries (☐): one or two lines each.

Registers: 30 landmark technologies, the timeline and awards. They are a sourced selection; the full lists grow on the website.

How to explore it

Step 1. Read Part 0, seven two-minute questions.

Step 2. Walk through your own home with Part 2.

Step 3. Pick a field you care about in Parts 4 and 5.

Step 4. Follow tags to music, films, cooking and crafts.

Step 5. Tick ☐ as you learn, and keep a maker's log.

Our promise

Knowledge belongs to the people who created it. Every fact here stands on a source listed in Part 12, and every "latest" fact says when it was checked. Where something could not be verified, we say so instead of guessing. Found a mistake? Tell us, and we will fix it and thank you.

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PART 0 — THE INTRODUCTION TWIG: HOW DID A SHARPENED STONE BECOME A SMARTPHONE?

About this Part. This is the doorway to the Technology section of the website, in the tree-twig format of the Curious Sparoows library: one Gateway page, seven Leaves and one Recap. Each Leaf answers one question in about two minutes (250 to 400 words), gives its short answer first and ends with a hook to the next question.

TEC-00 The Gateway: How did a sharpened stone become a smartphone?

Web address: /technology/stone-to-smartphone/ · **Keywords:** history of technology, how things work, types of technology, inventions timeline, technology for beginners

Technology is any tool or method people make to do something they could not do with bare hands: a stone flake that cuts, a wheel that rolls, a screw that holds, a chip that thinks. The phone in your pocket stands on about three million years of such steps, and on the work of millions of people whose names we will never know.

This short course asks one question in seven steps: **how does technology grow, and how do the things around us actually work?** It goes from the first stone tools to the wheel, the humble screw, electricity, the refrigerator and the internet, and ends with today's frontier. Each page takes about two minutes.

By the end you will be able to:

- Say what the first technology was and how old it is.
- Explain why the wheel came late.
- Describe how a screw turns a twist into a grip.
- Explain why electricity changed every other technology.
- Describe how a fridge moves heat out of a box.
- Trace a message from a server to your phone.
- Name today's frontier technologies and their limits.

TEC-00.01 What was the first technology?

> **Short answer:** The oldest known tools are sharp stone flakes from Lomekwi, Kenya, about 3.3 million years old, older than our genus *Homo*. Control of fire, at least a million years ago, was the second great technology. India's own stone tools at Attirampakkam (Tamil Nadu) go back about 1.5 million years.

Keywords: first tool ever, Lomekwi stone tools, Oldowan, Acheulean handaxe, prehistoric technology India

Stone that cuts

Striking one stone with another breaks off a flake with an edge sharper than teeth. The **Oldowan** tools (about 2.6 million years ago, Olduvai Gorge) and the teardrop-shaped **Acheulean handaxes** (from about 1.76 million years ago) were used for a million years and more.

Fire, clothing, containers

Fire gave warmth, protection and cooked food (see COK). Needles of bone (about 40,000 years ago or earlier) gave sewn clothing (SEW). Pottery (about 20,000 years ago in China) gave cooking pots and storage.

Why it matters

Each tool made the next one possible. Archaeologists describe this as *cumulative culture*: humans learn from others and add improvements, so technology grows across generations.

> **Remember this:** Technology is older than our species; it began with a sharp edge.

Next: If people had sharp tools so early, why did the wheel take so long?

TEC-00.02 Why did the wheel take so long to invent?

Short answer: The hard part was not the round wheel but the **axle**: a smooth pin that fits a hole in the wheel tightly enough to carry weight but loosely enough to turn. That needed metal tools and careful carpentry, so wheeled vehicles appear only about 3500 BCE, after farming, pottery and copper.

Keywords: invention of the wheel, history of the wheel, Ljubljana Marshes wheel, potter's wheel, wheel and axle

The evidence

- **Potter's wheel**: Mesopotamia, about 4000 BCE or earlier.
- **Wheeled vehicles**: pictures and models appear around 3500 to 3300 BCE from Mesopotamia to Central Europe; the Ljubljana Marshes wheel and axle (Slovenia) is about 5,150 years old.
- **India**: toy carts of clay from Harappan sites and real solid-wheeled bullock carts; the spoked wheel came later and became the *chakra* of the Konark Sun Temple and the national flag.

Why late?

- **Axle fit**: too tight and it jams; too loose and it wobbles.
- **Roads and animals**: wheels need flat ground and draught animals.
- **No need**: in the Americas, without horses or oxen, wheels appeared on toys but not carts.

Remember this: The wheel's genius is the axle, and that needed metal tools.

Next: Another simple machine hides in every piece of furniture. How does a screw hold?

TEC-00.03 How does a screw hold things together?

Short answer: A screw is an inclined plane wrapped around a cylinder. Turning it pushes it forward with great force, and the friction of its thread against the material holds it in place. That is why a small twist with a screwdriver can hold a heavy shelf.

Keywords: how screws work, screw simple machine, Archimedes screw, nut and bolt, fasteners

From pump to fastener

The screw as a machine is credited to Archimedes and others in the 3rd century BCE (the water-lifting Archimedes screw). Screw presses squeezed olives and grapes, and later printed books (TEC-08). Metal screws as fasteners became common only after machine-made threads in the 18th century.

Standard threads

Joseph Whitworth proposed a standard thread in 1841; later came metric (ISO) threads. Standards let a nut from one factory fit a bolt from another, a quiet revolution behind mass production.

Screws and their cousins

- **Wood screw**: cuts its own thread.
- **Bolt and nut**: clamp parts together.
- **Rivet**: a permanent pin (aircraft, bridges).
- **Nail**: holds by friction alone.

Remember this: A screw turns a gentle twist into a strong, lasting grip.

Next: Screws hold machines together. What made machines come alive?

TEC-00.04 Why did electricity change everything?

Short answer: Electricity turns energy from one place (a dam, a coal plant, a solar panel) into a form that travels instantly through wires and can become light, heat, motion, sound or information anywhere. Every later technology, from the fridge to the internet, depends on it.

Keywords: history of electricity, Faraday, Edison and Tesla, how electricity works, electrification of India

Key steps

- **1800**: Alessandro Volta's battery.
- **1831**: Michael Faraday discovers electromagnetic induction, the principle of every generator and transformer.
- **1879**: practical incandescent lamps by Thomas Edison and Joseph Swan.
- **1880s and 1890s**: the "current wars"; Nikola Tesla and George Westinghouse's alternating current (AC) wins for long-distance supply.
- **India**: Kolkata got electric street lighting in 1879 (demonstration) and public supply in 1899; Asia's early hydroelectric station at Sivasamudram, Karnataka (1902), powered the Kolar gold fields. India declared every village electrified in 2018.

What it unlocked

Lighting, motors, radio, refrigeration, computers and the grid itself. The US National Academy of Engineering ranked electrification the greatest engineering achievement of the 20th century (2000).

Remember this: Electricity is energy that can travel and transform.

Next: One of electricity's best gifts is cold. How does a fridge make cold?

TEC-00.05 How does a fridge make cold?

Short answer: A fridge does not make cold; it moves heat out. A liquid refrigerant evaporates in pipes inside the fridge, soaking up heat (just as sweat cools your skin). A compressor then squeezes the vapour, and the coils at the back release that heat into the room as the refrigerant turns back into liquid.

Keywords: how a refrigerator works, vapour compression cycle, refrigerant, compressor, history of refrigeration

The cycle

1. **Evaporator** (inside): low-pressure liquid boils and absorbs heat.
2. **Compressor**: squeezes the gas, making it hot.
3. **Condenser** (coils at the back): hot gas gives heat to the room and condenses.
4. **Expansion valve**: pressure drops, and the cycle repeats.

A short history

Jacob Perkins patented a vapour-compression machine (1834); household fridges spread in the 1920s and 1930s. CFC refrigerants harmed the ozone layer and were phased out under the Montreal Protocol (1987); the Kigali Amendment (2016) phases down HFCs, which warm the climate.

The same idea, elsewhere

Air conditioners and heat pumps use the same cycle. India's traditional cooling, the clay *matka* and *khus* screens, works by evaporation too.

Remember this: A fridge is a heat pump: it carries heat from inside to outside.

Next: From the kitchen to the world: how does the internet reach your phone?

TEC-00.06 How does the internet reach your phone?

Short answer: Your message is split into small packets with addresses. They travel by radio to a mobile tower or Wi-Fi router, then through fibre-optic cables (including undersea cables) and routers to a data centre, and the reply comes back the same way, all in a fraction of a second. Common rules (TCP/IP) let different networks talk.

Keywords: how the internet works, packet switching, TCP/IP, undersea cables, 5G India

The building blocks

- **Packet switching**: Paul Baran and Donald Davies (1960s); ARPANET (1969).
- **TCP/IP**: Vint Cerf and Bob Kahn (1974); the internet adopted it on 1 January 1983.
- **World Wide Web**: Tim Berners-Lee at CERN (1989 to 1991).
- **Fibre optics**: light in glass; Narinder Singh Kapany's pioneering work (1950s) and Charles Kao's Nobel-winning insight (2009 prize).
- **Undersea cables**: carry about 99 per cent of intercontinental data.

India online

Public internet arrived with VSNL on 15 August 1995. Cheap data after 2016, Aadhaar, UPI (2016) and 5G (from October 2022) made India one of the world's largest online populations.

Remember this: The internet is a network of networks sharing one set of rules.

Next: So what is new today, and what comes next?

TEC-00.07 What is the newest technology today?

Short answer: As of November 2026, the fastest-moving fields are generative artificial intelligence, AI chips and data centres, renewable energy and batteries, gene editing and mRNA medicine, quantum computing, reusable rockets and brain-computer interfaces. Each is real but also surrounded by hype: check claims against peer-reviewed or official sources.

Keywords: latest technology 2026, emerging technologies, generative AI, quantum computing, brain computer interface

Frontier fields (dated, November 2026)

- **AI**: large language models that write, code and reason; chip makers such as NVIDIA (its CEO Jensen Huang received the IEEE Medal of Honor, 2026).
- **Neural interfaces**: the 2026 Queen Elizabeth Prize for Engineering honoured pioneers of cochlear implants and brain-computer interfaces.
- **Light-controlled neurons**: the 2026 Nobel Prize in Physiology or Medicine honoured optogenetics.
- **Quantum information**: the 2025 Turing Award (announced 2026) went to Charles Bennett and Gilles Brassard for quantum cryptography.
- **Energy**: cheap solar and batteries; India's target of 500 GW of non-fossil capacity by 2030.
- **Space**: reusable rockets, India's Chandrayaan-3 landing near the Moon's south pole (2023) and its first space docking (SpaDeX, 2025).

How to judge "new"

Ask: does it work outside the lab, at what cost, for whom, and with what risks?

Remember this: The frontier moves every year; this page is dated so you know when it was true.

Next: The Recap ties the seven questions together.

TEC-00.08 Recap: from a stone edge to a thinking machine

- **TEC-00.01**: The first tools are 3.3 million years old.
- **TEC-00.02**: The wheel needed a good axle, and came late.
- **TEC-00.03**: A screw is a wrapped ramp that grips.
- **TEC-00.04**: Electricity carries energy anywhere and powers every later technology.
- **TEC-00.05**: A fridge pumps heat out.
- **TEC-00.06**: The internet sends addressed packets through networks that share rules.
- **TEC-00.07**: Today's frontier is AI, energy, biotech, quantum and space.

Where to go next: the Master Map (Part 1), then the devices in your own home (Part 2). **Quick quiz:** What are the six simple machines? Who discovered electromagnetic induction? What does TCP/IP do?

PART 1 — THE MASTER MAP OF TECHNOLOGY

About this Part. The whole hierarchy on one page. The brief asks for two trunks: **things** (devices, machines and tools) and **fields** (technology applied to a domain such as medicine or textiles). Every item has an address in one trunk and links to the other.

TEC-01 The hierarchy: from Forest to Leaf

Before you start — TEC-01. The Forest is "Technology". Trunk A, *Things*, splits into electronic devices, home appliances, mechanical devices and machines, and tools. Trunk B, *Fields*, splits into about 25 domains. A smartphone sits under Trunk A (handheld devices) and links to communication, computing and AI in Trunk B. **See also:** TEC-00 (the Gateway).

- ☐ **TEC-01.1 Trunk A, Things: electronic devices** (Part 2).
- ☐ **TEC-01.2 Trunk A, Things: home appliances** (Part 2).
- ☐ **TEC-01.3 Trunk A, Things: mechanical devices, machines and vehicles** (Part 3).
- ☐ **TEC-01.4 Trunk A, Things: tools and instruments** (Part 3).
- ☐ **TEC-01.5 Trunk B, Fields**: textiles, printing, art, construction, transport, energy, agriculture, food, water, medicine, communication, computing and AI, space, defence, materials, biotechnology, environment, finance, education, entertainment and assistive technology (Parts 4 and 5).
- ☐ **TEC-01.6 Registers**: 30 landmark technologies, the great timeline and awards (Part 6).
- ☐ **TEC-01.7 The human side**: people and institutions (Part 7), ideas (Part 8), the practical guide (Part 9) and reviews (Part 10).

PART 2 — TRUNK A: ELECTRONIC DEVICES AND HOME APPLIANCES

About this Part. The devices of daily life. Each line says what it is, the idea behind it, who or when it began, and where it stands now. "Now" means November 2026; specifications change fast, so the website pages will carry dated details from manufacturers.

TEC-02 Electronic devices

Before you start — TEC-02. Almost every electronic device rests on one invention: the transistor (Bell Labs, 1947, John Bardeen, Walter Brattain and William Shockley), and on the integrated circuit (Jack Kilby, 1958; Robert Noyce, 1959) that packs billions of transistors on a chip. Gordon Moore observed in 1965 that the number doubled regularly ("Moore's law"); the pace has slowed but not stopped.

☐ TEC-02.1 Handheld and wearable

- ☐ **TEC-02.1a Mobile phone**: Martin Cooper's Motorola call (1973); India's first mobile call on 31 July 1995 (Jyoti Basu to Sukh Ram); smartphones from IBM Simon (1994) to the iPhone (2007) and Android (2008); India is the world's second-largest smartphone market and a major manufacturer.
- ☐ **TEC-02.1b Tablet and e-reader**: touchscreen slates; Kindle (2007) used E Ink (electronic paper that reflects light like print).
- ☐ **TEC-02.1c Smartwatch and fitness band**: sensors for steps, heart rate and sleep; accuracy varies and they are not medical devices unless certified.
- ☐ **TEC-02.1d Wearables, AR and VR headsets**: smart glasses, mixed-reality headsets; still searching for everyday uses.
- ☐ **TEC-02.1e Calculator and pocket electronics**: Texas Instruments' handheld (1967 prototype); a cheap calculator holds more computing than Apollo's guidance computer.

☐ TEC-02.2 Computers

- ☐ **TEC-02.2a Desktop PC**: from the Altair (1975), Apple II (1977) and IBM PC (1981) to today's machines.
- ☐ **TEC-02.2b Laptop**: portable computers from the 1980s; India's affordable laptop schemes for students.
- ☐ **TEC-02.2c Servers and data centres**: buildings of computers that run the cloud and AI; huge users of electricity and water.
- ☐ **TEC-02.2d Supercomputers**: India's PARAM (C-DAC, 1991) and the National Supercomputing Mission; the world's fastest are listed twice a year on top500.org.
- ☐ **TEC-02.2e Single-board computers**: Raspberry Pi (2012) and Arduino microcontrollers (2005) for learning and making.
- ☐ **TEC-02.2f Parts of a computer**: CPU, GPU, memory (RAM), storage (hard disk, SSD), motherboard, power supply, input and output.

☐ TEC-02.3 Audio and display

- ☐ **TEC-02.3a Speakers and headphones**: a coil moves a cone (the moving-coil loudspeaker, 1925, Rice and Kellogg); noise cancellation adds opposite sound waves.
- ☐ **TEC-02.3b Microphones**: dynamic, condenser and the MEMS microphones in phones.
- ☐ **TEC-02.3c Television**: John Logie Baird's mechanical TV (1926), electronic TV (Philo Farnsworth, Vladimir Zworykin); Doordarshan began in Delhi on 15 September 1959; colour in India from 1982 (Asian Games).
- ☐ **TEC-02.3d Display technologies**: CRT, LCD, LED-backlit, OLED, micro-LED; resolutions from HD to 8K.
- ☐ **TEC-02.3e Projectors and monitors**: DLP and LCD projectors; high-refresh monitors for gaming.

☐ TEC-02.4 Networking

☐ **TEC-02.4a Modem and router**: the modem translates the line signal; the router directs traffic in the home.

☐ **TEC-02.4b Wi-Fi**: based on IEEE 802.11 (1997); CSIRO of Australia's patent was key; versions up to Wi-Fi 7.

☐ **TEC-02.4c Mobile networks**: 1G analogue to 5G; India's indigenous 4G and 5G stack by C-DOT and partners (BSNL rollout from 2024 to 2025).

☐ **TEC-02.4d Fibre optics and switches**: light pulses in glass carry the internet's backbone (TEC-00.06); BharatNet aims to connect every gram panchayat.

☐ **TEC-02.4e Satellite internet**: low-Earth-orbit constellations; licensing in India under the Telecommunications Act (2023).

☐ TEC-02.5 Power

☐ **TEC-02.5a Batteries**: Volta (1800), lead-acid (Gaston Planté, 1859), lithium-ion (Nobel in Chemistry 2019 to John Goodenough, Stanley Whittingham and Akira Yoshino); sodium-ion and solid-state are next.

☐ **TEC-02.5b Chargers and power banks**: USB-C is now the common standard (India moved new phones to it from 2025, in line with the EU).

☐ **TEC-02.5c UPS and inverters**: backup power, common in Indian homes.

☐ **TEC-02.5d Solar inverters and rooftop solar**: PM Surya Ghar Muft Bijli Yojana (2024) subsidises rooftop solar.

☐ TEC-02.6 Cameras, drones, consoles and the smart home

☐ **TEC-02.6a Cameras**: from the camera obscura (described by Ibn al-Haytham, about 1021) and photography (Niépce, Daguerre, 1820s to 1839) to the digital sensor (CCD, Boyle and Smith, 1969; Nobel 2009) and the CMOS sensors in phones.

☐ **TEC-02.6b Drones**: remote and autonomous aircraft for photography, mapping, farming (Namo Drone Didi scheme) and delivery; India's Drone Rules (2021).

☐ **TEC-02.6c Gaming consoles**: Magnavox Odyssey (1972) to today's consoles; see SPT for esports.

☐ **TEC-02.6d Smart home and IoT**: connected lights, plugs, cameras, speakers; the Matter standard (2022) for compatibility; privacy and security risks.

TEC-03 Home appliances

Before you start — TEC-03. Appliances changed daily life, especially women's unpaid work, more than many famous inventions; the economist Ha-Joon Chang argued that the washing machine changed the world more than the internet (*23 Things They Don't Tell You about Capitalism*, 2010). India's Bureau of Energy Efficiency (BEE) star labels (from 2006) help compare running costs.

☐ **TEC-03.1 Refrigerator**: vapour-compression cycle (TEC-00.05); frost-free and inverter models; BEE star ratings.

☐ **TEC-03.2 Microwave oven**: Percy Spencer noticed a melted chocolate bar near a radar magnetron (1945); microwaves shake water molecules to heat food.

☐ **TEC-03.3 Oven, OTG and air fryer**: hot air and radiant heat; the air fryer is a small, fast convection oven.

☐ **TEC-03.4 Induction cooktop**: a changing magnetic field heats an iron pan directly; efficient and safe, but needs magnetic cookware.

☐ **TEC-03.5 Mixer-grinder**: India's kitchen workhorse; Sumeet (1963, designed by Satya Prakash Mathur) made it a household name for Indian wet grinding.

☐ **TEC-03.6 Pressure cooker**: see COK-10.3c.

☐ **TEC-03.7 Washing machine**: drum machines from the early 1900s; semi-automatic and fully automatic types; front- and top-loading.

☐ **TEC-03.8 Air conditioner**: Willis Carrier (1902); inverter ACs and star ratings; refrigerants with lower warming potential.

☐ **TEC-03.9 Air cooler**: evaporative cooling; works best in dry heat (Rajasthan), poorly in humid coasts.

☐ **TEC-03.10 Ceiling fan**: Philip Diehl (1882); BLDC (brushless) fans use far less power.

☐ **TEC-03.11 Water purifier**: filters, UV and RO (reverse osmosis); RO wastes water and removes minerals, so use it only where the water needs it (check TDS).

☐ **TEC-03.12 Geyser and water heater**: storage, instant and solar water heaters.

☐ **TEC-03.13 Vacuum cleaner and robot vacuum**: Hubert Cecil Booth's powered vacuum (1901); robot vacuums map rooms with sensors.

☐ **TEC-03.14 Electric iron**: Henry Seeley's patent (1882); steam irons.

☐ **TEC-03.15 Lighting**: incandescent, CFL, LED; India's UJALA scheme (2015) distributed over 36 crore LED bulbs.

PART 3 — TRUNK A: MECHANICAL DEVICES, MACHINES, VEHICLES AND TOOLS

About this Part. From the smallest fastener to the largest machines ever built. Mechanical technology is older than electronics and still holds the world together.

TEC-04 Fasteners and simple machines

Before you start — TEC-04. Greek thinkers (Archimedes, Hero of Alexandria) listed the simple machines; Renaissance scientists settled on six. Each trades force for distance: you push less but farther. **See also:** TEC-00.03.

- ☐ **TEC-04.1 Lever**: a bar on a pivot (fulcrum); seesaw, crowbar, scissors, the *dhenkli* well-sweep of Indian villages. "Give me a place to stand and I will move the Earth" (attributed to Archimedes).
- ☐ **TEC-04.2 Wheel and axle**: TEC-00.02; door knobs, steering wheels, the Persian wheel (*rahat*) for lifting water.
- ☐ **TEC-04.3 Pulley**: a wheel with a rope; well pulleys, cranes, lifts.
- ☐ **TEC-04.4 Inclined plane**: a ramp; how pyramids and temple towers were likely raised.
- ☐ **TEC-04.5 Wedge**: two inclined planes back to back; axe, knife, chisel, plough.
- ☐ **TEC-04.6 Screw**: an inclined plane wrapped around a cylinder (TEC-00.03).
- ☐ **TEC-04.7 Fasteners**: screws, nuts and bolts, nails (hand-forged, then machine-cut and wire nails), rivets, pins, washers, clips, zippers (Whitcomb Judson, 1893; Gideon Sundback's modern zip, 1913), Velcro (George de Mestral, 1955).
- ☐ **TEC-04.8 Gears, cams, springs and bearings**: gears transmit turning force (the Antikythera mechanism, about 100 BCE, is the oldest known geared device); ball bearings reduce friction.
- ☐ **TEC-04.9 Locks and keys**: Egyptian wooden pin locks (about 4,000 years ago), Linus Yale Jr's pin-tumbler lock (1860s), Aligarh's lock industry, today's smart locks.
- ☐ **TEC-04.10 Clocks**: sundials and water clocks (*ghati* in India); the mechanical escapement (China's Su Song, 1090s; European clocks, 14th century); Christiaan Huygens's pendulum clock (1656); quartz (1927) and atomic clocks; India's time is kept by CSIR-NPL, New Delhi.

TEC-05 Engines, vehicles and giant machines

- ☐ **TEC-05.1 Steam engine**: Thomas Newcomen (1712), James Watt's separate condenser (1769); the power of the Industrial Revolution.
- ☐ **TEC-05.2 Internal combustion engine**: Nikolaus Otto's four-stroke (1876), Rudolf Diesel (1890s); now being replaced in many uses by electric motors.
- ☐ **TEC-05.3 Electric motor**: Faraday's rotation (1821); motors drive fans, trains, EVs and robots.
- ☐ **TEC-05.4 Turbines and jet engines**: steam turbine (Charles Parsons, 1884), jet engines (Frank Whittle and Hans von Ohain, late 1930s).
- ☐ **TEC-05.5 Bicycle**: Karl Drais's running machine (1817), the chain-driven "safety bicycle" (John Kemp Starley, 1885); India's Atlas and Hero made it the people's vehicle.
- ☐ **TEC-05.6 Cars**: Karl Benz's Motorwagen (1885 to 1886); Ford's Model T assembly line (1913); Maruti 800 (1983) put India on wheels; electric vehicles and the FAME and PM E-DRIVE schemes.
- ☐ **TEC-05.7 Trains**: Stephenson's *Rocket* (1829); India's first passenger train, Bori Bunder to Thane (16 April 1853); Vande Bharat trains (2019) built at ICF Chennai; metros in many cities; the Mumbai–Ahmedabad high-speed rail under construction.
- ☐ **TEC-05.8 Ships and submarines**: dhows and the Indian Ocean trade, steamships, container ships (Malcolm McLean, 1956); India's indigenous aircraft carrier INS *Vikrant* (commissioned 2022).
- ☐ **TEC-05.9 Aircraft**: the Wright brothers (17 December 1903); India's first commercial flight, J. R. D. Tata from Karachi to Bombay (15 October 1932); HAL's Tejas light combat aircraft.

☐ **TEC-05.10 Rockets**: Tipu Sultan's iron-cased Mysorean rockets (1780s and 1790s) inspired Britain's Congreve rockets; Konstantin Tsiolkovsky, Robert Goddard (first liquid-fuel rocket, 1926); India's first rocket launch from Thumba (21 November 1963); reusable boosters today.

☐ **TEC-05.11 Cranes, excavators and tunnel-boring machines**: from Roman treadwheel cranes to TBMs that dug the Gotthard Base Tunnel (57 km, 2016) and India's Atal Tunnel (9 km, 2020, built by drill and blast).

☐ **TEC-05.12 The largest machines ever built**: the Large Hadron Collider (27 km ring, CERN, 2008); the Bagger 293 bucket-wheel excavator; container ships of about 400 m; the International Space Station (from 1998).

TEC-06 Tools and instruments

☐ **TEC-06.1 Hand tools**: hammer, screwdriver, spanner (wrench), pliers, saw, chisel, file, drill; the Indian *basula* (adze) and *randa* (plane) of the carpenter.

☐ **TEC-06.2 Power tools**: electric drill (Wilhelm Fein, 1895; Black & Decker's pistol grip, 1917), angle grinder, circular saw, jigsaw, cordless tools.

☐ **TEC-06.3 Measuring instruments**: ruler, vernier caliper (Pierre Vernier, 1631), micrometer, spirit level, multimeter, thermometer (Galileo's thermoscope; Fahrenheit and Celsius scales), barometer (Torricelli, 1643), weighing scales; the SI units (redefined in 2019).

☐ **TEC-06.4 Scientific instruments**: microscope (Leeuwenhoek's single-lens microscopes, 1670s), telescope (Galileo, 1609), spectroscope; India's observatories (Jantar Mantar, Jaipur, UNESCO 2010).

☐ **TEC-06.5 Farm tools**: plough, sickle, hoe (*kudal*), *khurpi*, seed drill (the Babylonian seed plough; India's bamboo seed drill, *tifan*; Jethro Tull's drill, 1701), threshers, tractors (India is the world's largest tractor maker by volume).

☐ **TEC-06.6 Craft tools**: the loom, spinning wheel (*charkha*), potter's wheel, needles and hooks (CRO, KNT, EMB, SEW).

PART 4 — TRUNK B: TECHNOLOGY BY FIELD (1)

About this Part. Technology applied to a domain. Each field lists its main branches, its landmark inventions and India's contribution. The brief names many fields; they are grouped into two Parts.

TEC-07 Textiles and clothing technology

☐ **TEC-07.1 Fibres**: natural (cotton, first spun in the Indus valley; silk, China; wool; jute; linen), regenerated (rayon, viscose) and synthetic (nylon, Wallace Carothers at DuPont, 1935; polyester, 1941).

☐ **TEC-07.2 Spinning**: hand spindle, *charkha*, the spinning jenny (Hargreaves, 1764), water frame (Arkwright, 1769), mule (Crompton, 1779), ring and rotor spinning.

☐ **TEC-07.3 Weaving**: handlooms, the flying shuttle (John Kay, 1733), power loom (Cartwright, 1785), the Jacquard loom (1804), whose punched cards inspired early computers.

☐ **TEC-07.4 Dyeing and printing**: indigo, madder, the Indian chintz and *kalamkari* traditions that astonished Europe; synthetic mauve (William Perkin, 1856).

☐ **TEC-07.5 Smart and technical textiles**: conductive and sensor fabrics, fire-resistant and medical textiles; India's National Technical Textiles Mission (2020).

☐ **TEC-07.6 The sewing machine and knitting machine**: Elias Howe (1846) and Isaac Singer (1851); William Lee's stocking frame (1589). See SEW and KNT.

TEC-08 Printing, books and art technology

☐ **TEC-08.1 Writing materials**: clay, palm leaf (Indian manuscripts), birch bark, papyrus, parchment, paper (Cai Lun, China, 105 CE, by tradition).

☐ **TEC-08.2 Printing**: woodblock printing (China; the *Diamond Sutra*, 868 CE), movable type (Bi Sheng, 1040s; Korea's metal type), Gutenberg's press (about 1450); India's first press at Goa (1556); Serampore Mission Press (1800).

☐ **TEC-08.3 Modern printing**: lithography, offset, digital printing, 3D printing (Charles Hull's stereolithography, 1984).

☐ **TEC-08.4 Art technology**: pigments (ochre, Indian yellow, lapis ultramarine), oil paint, paint tubes (John Rand, 1841), canvas, photography, digital art and drawing tablets, generative AI images (with debates about artists' consent and copyright). See BOK and MOV.

TEC-09 Construction, transport and energy

☐ **TEC-09.1 Construction materials**: mud brick, fired brick (Harappan standardised bricks), lime mortar, Roman concrete, iron and steel (Delhi's rust-resistant iron pillar, about 400 CE), Portland cement (1824), reinforced concrete.

☐ **TEC-09.2 Architecture and structures**: arches, domes and trusses; stepwells; skyscrapers (Home Insurance Building, Chicago, 1885); bridges (India's Chenab rail bridge, the world's highest railway arch bridge, opened 2025).

☐ **TEC-09.3 Transport systems**: roads and highways (Grand Trunk Road; the Golden Quadrilateral, 2001 to 2012), railways, ports, airports, traffic signals and FASTag.

☐ **TEC-09.4 Energy**: firewood to coal, oil, gas, nuclear (India's three-stage programme of Homi Bhabha; Tarapur, 1969), hydro, solar (Bhadla Solar Park, Rajasthan), wind, green hydrogen (National Green Hydrogen Mission, 2023), the grid (One Nation One Grid, 2013).

TEC-10 Agriculture, food, water and sanitation

☐ **TEC-10.1 Agriculture technology**: irrigation, the plough, the Green Revolution's high-yield seeds, tractors, drip irrigation, soil health cards, drones, precision farming. See GRD.

☐ **TEC-10.2 Food technology**: preservation, canning, refrigeration, pasteurisation (Louis Pasteur, 1860s), the cold chain, milk processing (Amul and Operation Flood), packaging, food fortification. See COK.

☐ **TEC-10.3 Water**: stepwells and tanks, dams, canals, water treatment (chlorination from 1900s), desalination, Jal Jeevan Mission (2019) tap connections.

☐ **TEC-10.4 Sanitation**: Harappan drains (among the world's earliest), the flush toilet (John Harington, 1596; Alexander Cumming's S-bend, 1775), sewers, twin-pit toilets, Swachh Bharat Mission (2014); Bindeshwar Pathak's Sulabh toilets (1970).

PART 5 — TRUNK B: TECHNOLOGY BY FIELD (2)

TEC-11 Medical technology

Before you start — TEC-11. Medical technology is the long list the brief asks for. Every medical claim on the website will cite peer-reviewed reviews or regulators (WHO, CDSCO, US FDA). **This index gives information, not medical advice.**

- ☐ **TEC-11.1 Diagnostics**: stethoscope (René Laennec, 1816), thermometer, blood pressure cuff (Riva-Rocci, 1896), lab tests, rapid tests, ECG (Willem Einthoven, Nobel 1924), pulse oximeter (Takuo Aoyagi, 1970s), point-of-care devices.
- ☐ **TEC-11.2 Imaging**: X-rays (Wilhelm Röntgen, 1895; first Nobel in Physics, 1901), ultrasound, CT (Godfrey Hounsfield and Allan Cormack, Nobel 1979), MRI (Paul Lauterbur and Peter Mansfield, Nobel 2003), PET, AI-assisted reading of scans. *Note:* India's PCPNDT Act bans sex determination by ultrasound.
- ☐ **TEC-11.3 Surgery**: anaesthesia (1846), antisepsis (Joseph Lister, 1867), Sushruta's ancient descriptions of rhinoplasty, laparoscopic (keyhole) surgery, robotic surgery, lasers.
- ☐ **TEC-11.4 Prosthetics and implants**: the Jaipur Foot (P. K. Sethi and Ram Chandra Sharma, 1968), artificial joints, pacemakers (1958), cochlear implants (QEPrize 2026), stents, brain-computer interfaces.
- ☐ **TEC-11.5 Pharmaceuticals**: aspirin (1897), insulin (1921 to 1922), penicillin (Fleming, 1928; mass-produced in the 1940s); India as the "pharmacy of the world" for generic medicines.
- ☐ **TEC-11.6 Vaccines**: smallpox vaccination (Edward Jenner, 1796; variolation earlier in India, China and Africa), polio vaccines (Salk, Sabin), mRNA vaccines (Katalin Karikó and Drew Weissman, Nobel 2023); India's Covaxin and Covishield; Serum Institute of India, the world's largest vaccine maker by doses.
- ☐ **TEC-11.7 Telemedicine and digital health**: eSanjeevani (2019), the Ayushman Bharat Digital Mission (2021), CoWIN.
- ☐ **TEC-11.8 Medical devices for hospitals**: ventilators, dialysis machines (Willem Kolff, 1943), incubators, oxygen concentrators, radiotherapy.
- ☐ **TEC-11.9 Neurotechnology**: EEG, deep brain stimulation, optogenetics (Nobel 2026, Karl Deisseroth, Peter Hegemann and Georg Nagel).

TEC-12 Communication, computing and AI

- ☐ **TEC-12.1 Communication**: drums and smoke signals, postal systems (India Post, 1854), the telegraph (Samuel Morse, 1837 to 1844; India's Calcutta–Diamond Harbour line, 1851), telephone (Alexander Graham Bell, 1876; Antonio Meucci earlier), radio (Marconi; J. C. Bose's millimetre-wave work, 1895), television, satellites, the internet, mobile.
- ☐ **TEC-12.2 Computing**: the abacus, Indian decimal place value and zero (Brahmagupta, 628 CE), Charles Babbage's engines and Ada Lovelace's notes (1843), Alan Turing (1936), ENIAC (1945), the transistor, the microprocessor (Intel 4004, 1971), personal computers, the cloud.
- ☐ **TEC-12.3 Software**: operating systems (Unix, 1969; Linux, 1991), programming languages (FORTRAN, C, Python), open source, India's IT services industry (TCS, 1968; Infosys, 1981; Wipro).
- ☐ **TEC-12.4 Artificial intelligence**: the term coined at Dartmouth (1956); expert systems; machine learning; deep learning (the 2018 Turing Award to Yoshua Bengio, Geoffrey Hinton and Yann LeCun; Hinton and John Hopfield's Nobel in Physics, 2024); AlphaFold (Nobel in Chemistry 2024 to Demis Hassabis and John Jumper, with David Baker); large language models and generative AI; the IndiaAI Mission (2024).
- ☐ **TEC-12.5 Quantum computing and quantum information**: qubits, quantum cryptography (Turing Award 2025 to Charles Bennett and Gilles Brassard); India's National Quantum Mission (2023).
- ☐ **TEC-12.6 Cybersecurity**: passwords and two-factor authentication, encryption, CERT-In, the Digital Personal Data Protection Act (2023).

TEC-13 Space, defence and materials

- ☐ **TEC-13.1 Space**: Sputnik (1957), Gagarin (1961), Apollo 11 (1969); ISRO (1969) under Vikram Sarabhai; Aryabhata (1975), SLV-3 (1980), INSAT, PSLV, Chandrayaan-1 (2008, water on the Moon), Mangalyaan (2014, Mars orbit on the first attempt), Chandrayaan-3 (2023), Aditya-L1 (2024), SpaDeX docking (2025), Shubhanshu Shukla on the International Space Station (Axiom-4, 2025); Gaganyaan human spaceflight in preparation; NavIC navigation.

☐ **TEC-13.2 Defence technology**: from bows and *wootz* steel to gunpowder, radar, missiles; India's DRDO (1958), the Integrated Guided Missile Development Programme under A. P. J. Abdul Kalam (1983), BrahMos, Agni, Tejas, Akash. Presented as history and engineering, not as weapons advice.

☐ **TEC-13.3 Materials**: copper, bronze and iron ages; *wootz* (crucible) steel of South India, prized as "Damascus steel"; plastics (Bakelite, 1907); semiconductors (silicon); composites, graphene (isolated 2004; Nobel 2010), superalloys; India's semiconductor mission (2021) and first chip fabs.

TEC-14 Biotechnology, environment, finance, education, entertainment and assistive technology

☐ **TEC-14.1 Biotechnology**: fermentation, the DNA double helix (1953), recombinant DNA (1973), PCR (Kary Mullis, 1983), the Human Genome Project (2003), CRISPR gene editing (Jennifer Doudna and Emmanuelle Charpentier, Nobel 2020); India's Bt cotton (2002), Genome India project, Biocon.

☐ **TEC-14.2 Environmental technology**: pollution control, catalytic converters, air-quality monitors, waste-to-energy, recycling, carbon capture, e-waste processing; India's E-Waste (Management) Rules (2022).

☐ **TEC-14.3 Financial technology**: coins (India's punch-marked coins), paper money, banks, ATMs (1967; India's first, HSBC Mumbai, 1987), credit cards, online banking, UPI (NPCI, 2016; over 20 billion transactions a month by 2025), RuPay, digital rupee pilots, Aadhaar-enabled payments; risks: fraud and scams.

☐ **TEC-14.4 Education technology**: the blackboard, printed textbooks, radio and TV classes (SITE 1975), MOOCs, SWAYAM (2017), DIKSHA, AI tutors; the digital divide.

☐ **TEC-14.5 Entertainment technology**: cinema (MOV), recorded sound (MUS), radio, TV, video games (SPT), streaming, VR.

☐ **TEC-14.6 Assistive technology**: Braille (Louis Braille, 1824; Bharati Braille for Indian scripts), white canes, hearing aids, screen readers, wheelchairs, sign-language apps; India's Accessible India Campaign (2015) and ADIP scheme.

PART 6 — REGISTERS

About this Part. Three registers: 30 landmark technologies, the great timeline from stone tools to AI, and the major awards with their latest results (checked 13 November 2026). All are a sourced selection; the full lists grow on the website.

TEC-15 Landmark technologies (register of 30)

Before you start — TEC-15. The brief's register has seven fields. To fit the page it is split into two linked tables.

☐ TEC-15.1 Card A: origin

Item	Category path	First idea / inventor / year	Key milestones	How it works (one line)
Stone tools	Things › Tools	Unknown hominins, Lomekwi, c. 3.3 million years ago	Oldowan, Acheulean, microliths	A struck stone breaks into a sharp edge
Control of fire	Fields › Energy	Early *Homo*, at least 1 million years ago	Hearths, kilns, furnaces	Combustion releases stored chemical energy
Wheel and axle	Things › Simple machines	Mesopotamia and Europe, c. 3500 BCE	Spokes, bearings, tyres	A turning wheel on a fixed axle reduces friction
Plough	Fields › Agriculture	Mesopotamia and the Indus, c. 4th millennium BCE	Iron share, steel plough (John Deere, 1837), tractor	A wedge breaks and turns soil
Writing	Fields › Communication	Sumer, c. 3200 BCE; Indus script undeciphered	Alphabets, paper, printing	Symbols store language outside memory
Iron smelting	Fields › Materials	Anatolia and beyond, c. 1200 BCE; India early	Wootz steel, Bessemer process (1856)	Heat and carbon pull oxygen from iron ore
Decimal place value and zero	Fields › Computing	India, first centuries CE; Brahmagupta's rules (628)	Arabic transmission, Fibonacci (1202)	Ten digits whose value depends on position
Paper	Fields › Printing	Cai Lun, China, 105 CE (by tradition)	Samarkand, Baghdad, Europe, India	Matted plant fibres dried into sheets
Printing press	Fields › Printing	Bi Sheng (1040s); Gutenberg (c. 1450)	Steam press, offset, digital	Inked type pressed onto paper
Magnetic compass	Things › Instruments	China, 11th century	Gyrocompass, GPS	A magnetised needle aligns with Earth's field
Mechanical clock	Things › Mechanical	Su Song (1090s); Europe (14th century)	Pendulum, quartz, atomic	An escapement releases energy in equal ticks
Steam engine	Things › Engines	Newcomen (1712); Watt (1769)	Locomotives, steamships, turbines	Steam pressure pushes a piston
Electric battery	Things › Power	Volta (1800)	Lead-acid, lithium-ion	Chemical reactions push electrons through a circuit
Electric generator	Fields › Energy	Faraday (1831)	Power stations, grids	A moving magnet induces current in a coil
Telegraph	Fields › Communication	Morse and Vail (1837 to 1844)	Undersea cable (1866), telex	Pulses of current carry coded letters
Telephone	Fields › Communication	Bell (1876)	Exchanges, mobile phones	A microphone turns voice into electrical signals
Light bulb	Things › Home	Swan and Edison (1879)	Fluorescent, LED	Current heats a filament, or excites a semiconductor
Internal combustion engine	Things › Engines	Otto (1876); Diesel (1890s)	Cars, aircraft, generators	Fuel burns inside a cylinder to push a piston
Radio	Fields › Communication	Marconi, Bose and others (1890s)	Broadcasting, Wi-Fi, mobile	Electromagnetic waves carry signals through air
Aeroplane	Things › Vehicles	Wright brothers (1903)	Jet age, wide-body airliners	Wings create lift as air flows over them
Antibiotics	Fields › Medicine	Fleming (1928); Florey and	Many antibiotic classes;	Chemicals kill or stop bacteria

		Chain (1940s)	resistance	
Television	Things › Display	Baird (1926), Farnsworth (1927)	Colour, digital, smart TV	Images are scanned, sent and redrawn as signals
Nuclear power	Fields › Energy	Fermi's reactor (1942); first grid power (1954)	Tarapur (1969), fast breeders	Splitting atoms releases heat to make steam
Transistor	Things › Electronics	Bardeen, Brattain and Shockley (1947)	Integrated circuit, microprocessor	A small current controls a larger one
Satellite	Fields › Space	Sputnik (1957); Aryabhata (1975)	GPS, INSAT, NavIC	An object falls around Earth without hitting it
Laser	Things › Instruments	Theodore Maiman (1960)	Fibre optics, surgery, barcode	Atoms emit light in step, making a tight beam
Personal computer	Things › Computers	Altair (1975), Apple II (1977), IBM PC (1981)	Laptops, smartphones	A microprocessor runs stored programs
Internet and Web	Fields › Communication	ARPANET (1969); TCP/IP (1974); Web (1989 to 1991)	Broadband, mobile internet, cloud	Packets travel under shared rules
Smartphone	Things › Handheld	IBM Simon (1994); iPhone (2007)	App stores, 4G, 5G	A pocket computer with radios and sensors
Generative AI	Fields › Computing and AI	Transformer architecture (Vaswani and colleagues, 2017)	Large language models, image and video models	Neural networks learn patterns to predict and generate

☐ TEC-15.2 Card B: state of the art and India

Item	Current state of the art (Nov 2026)	Indian contribution
Stone tools	Studied by archaeology; inspire materials science	Attirampakkam tools, c. 1.5 million years old; microliths of Bhimbetka
Control of fire	Clean cooking (LPG, induction) replaces smoky fuels	Ujjwala Yojana (2016) LPG connections
Wheel and axle	Airless tyres, magnetic levitation	Konark's stone wheels; Indian Railways' wheel factories
Plough	Precision and no-till farming	Iron ploughs of ancient India; tractor manufacturing
Writing	Digital text and Unicode	Brahmi and its many daughter scripts; Unicode support for Indian scripts
Iron smelting	Green steel with hydrogen	Wootz steel; the Delhi iron pillar; Tata Steel (1907)
Decimal place value and zero	Basis of all computing	The invention itself; Aryabhata, Brahmagupta, the Bakhshali manuscript
Paper	Recycled and digital alternatives	Handmade paper of Sanganer (Jaipur) and Kalpi
Printing press	Digital and 3D printing	Goa press (1556); Serampore; Indian-language typefaces
Magnetic compass	Satellite navigation	NavIC regional navigation system
Mechanical clock	Optical atomic clocks	Jantar Mantar instruments; Indian Standard Time from CSIR-NPL
Steam engine	Mostly heritage; turbines remain	Darjeeling Himalayan Railway (UNESCO)
Electric battery	Lithium-ion, sodium-ion, solid-state research	Battery manufacturing PLI scheme; lithium finds in Jammu and Kashmir (2023)
Electric generator	Huge wind and hydro turbines	Sivasamudram (1902); BHEL
Telegraph	Retired; India's last telegram was sent on 14 July 2013	Calcutta–Diamond Harbour line (1851)
Telephone	Mobile and internet calling	C-DOT exchanges (Sam Pitroda, 1984)
Light bulb	LEDs	UJALA LED programme
Internal combustion engine	Being replaced by electric drives in many uses	Two-wheeler giants; ethanol blending (E20, 2025)
Radio	Software-defined and 5G radios	J. C. Bose's microwave experiments; All India Radio (1936)
Aeroplane	Fuel-efficient jets, electric aircraft research	HAL; Tata's first airline; drones

Antibiotics	Fighting antimicrobial resistance	Generic antibiotics for the world; Kerala's AMR action plan
Television	OLED and streaming	Doordarshan; SITE (1975) educational TV
Nuclear power	Small modular reactors, fusion research (ITER)	Three-stage programme; thorium research; ITER partner
Transistor	Chips with billions of transistors at a few nanometres	Semiconductor mission; chip design talent
Satellite	Mega-constellations	ISRO's low-cost launches; 104 satellites on one PSLV (2017)
Laser	Ultrafast lasers (Nobel 2018, 2023)	RRCAT Indore laser research
Personal computer	AI PCs, ARM chips	PARAM supercomputers; IT services
Internet and Web	5G, fibre, satellite internet	UPI, Aadhaar, the "India Stack"
Smartphone	Foldables, on-device AI	Large-scale phone assembly in India
Generative AI	Multimodal models and AI agents	IndiaAI Mission; Indian-language models such as BharatGen and Sarvam

TEC-16 The great timeline (year-wise register)

Before you start — TEC-16. The brief's chain: stone tools, fire, wheel, metals, writing, printing, steam, electricity, electronics, computers, internet, mobile, AI. Each device also gets its own timeline on its website page.

Year	Event	Where	Why it matters
c. 3.3 million years ago	Lomekwi stone tools	Kenya	Oldest known technology
c. 1.5 million years ago	Acheulean tools at Attirampakkam	Tamil Nadu	Early technology in India
c. 1 million years ago	Controlled fire (Wonderwerk Cave)	South Africa	Heat, light, cooking
c. 9500 BCE	Farming	Fertile Crescent	Settled life
c. 3500 BCE	Wheel and axle; copper smelting	Mesopotamia, Europe	Transport and metals
c. 3200 BCE	Writing	Sumer, Egypt	Recorded knowledge
c. 2600 BCE	Planned Harappan cities with drains and standard bricks	Indus valley	Urban engineering
c. 1200 BCE	Iron Age spreads	Anatolia, India	Stronger tools
105 CE	Paper (by tradition)	China	Cheap writing material
c. 400 CE	Delhi iron pillar	India	Rust-resistant iron
628	Brahmagupta's rules for zero	India	Mathematics of computing
c. 1450	Gutenberg's press	Germany	Mass printing
1712	Newcomen steam engine	England	Industrial power
1769	Watt's improved engine	Scotland	Industrial Revolution
1800	Volta's battery	Italy	Steady electric current
1831	Faraday's induction	England	Generators and transformers
1853	First passenger train in India	Mumbai to Thane	Railways in India
1876	Telephone	USA	Voice at a distance
1879	Practical light bulb	UK and USA	Electric lighting
1895	Radio waves (Marconi, Bose)	Italy, India	Wireless communication
1903	Powered flight	USA	The air age
1947	Transistor	USA	Electronics
1957	Sputnik	USSR	Space age
1963	First rocket from Thumba	Kerala	India's space programme
1969	ARPANET; Moon landing; ISRO founded	USA, Moon, India	Networks and space
1971	Microprocessor	USA	Computers on a chip
1975	Aryabhata satellite	India	India's first satellite
1983	Internet adopts TCP/IP; first	USA	Internet and mobile

	commercial mobile phone service in the USA		
1991	World Wide Web public	CERN	The Web
1995	Internet and mobile service in India	India	Digital India begins
2007	iPhone	USA	Smartphone era
2016	UPI launched	India	Instant digital payments
2017	Transformer neural network	USA	Basis of today's AI
2022	5G in India; generative AI chatbots go mainstream	India, world	Fast mobile and public AI
2023	Chandrayaan-3 lands near the lunar south pole	Moon	India's lunar landing
2025	SpaDeX docking; Shubhanshu Shukla becomes the first Indian aboard the ISS (Axiom-4), and the first Indian in space since Rakesh Sharma (1984)	India, orbit	Space capability
2026	Nobel prizes for optogenetics (medicine) and the IceCube neutrino observatory (physics)	Stockholm	Latest science-technology honours

TEC-17 Awards for technology (register)

Award	Body	Since	What it honours	Latest (checked 13 Nov 2026)	Note
Nobel Prize in Physics	Royal Swedish Academy of Sciences	1901	Discoveries and inventions in physics	2026: Francis Halzen (IceCube Neutrino Observatory)	C. V. Raman, 1930, India's first science Nobel
Nobel Prize in Chemistry	Royal Swedish Academy of Sciences	1901	Chemistry, including materials and biotech	2026: Henri Kagan and Kenso Soai (non-linear effects and autocatalysis in asymmetric synthesis)	Venkatraman Ramakrishnan, 2009 (born in India)
Nobel Prize in Physiology or Medicine	Nobel Assembly, Karolinska Institutet	1901	Medical discoveries	2026: Karl Deisseroth, Peter Hegemann and Georg Nagel (optogenetics)	Har Gobind Khorana, 1968 (born in undivided India)
ACM A. M. Turing Award	Association for Computing Machinery	1966	Lasting contributions to computing	2025 (announced March 2026): Charles Bennett and Gilles Brassard (quantum cryptography)	Raj Reddy, 1994, first of Asian origin
IEEE Medal of Honor	IEEE	1917	Exceptional contributions to electrical and electronic technology	2026: Jensen Huang (accelerated computing)	IEEE's highest award
Millennium Technology Prize	Technology Academy Finland	2004	Innovations for a better life	2024: B. Jayant Baliga (IGBT power transistor); next in 2027	Biennial
Queen Elizabeth Prize for Engineering	QEPrize Foundation	2013	Engineering that benefits humanity	2026: pioneers of modern neural interfaces, including Graeme Clark, Ingeborg and Erwin Hochmair, Blake Wilson and John Donoghue	£500,000 prize
Charles Stark Draper	US National Academy	1989	Engineering	See nae.edu	Not verified for this

Prize	of Engineering		achievement		edition
Rashtriya Vigyan Puraskar	Government of India	2024 (replacing the Shanti Swarup Bhatnagar Prize and others)	Vigyan Ratna, Vigyan Shri, Vigyan Yuva (Shanti Swarup Bhatnagar) and Vigyan Team	2025: Vigyan Ratna to Jayant Narlikar (posthumous)	Announced in October 2025
National Technology Day awards	Technology Development Board	1999 (Technology Day itself marks Pokhran-II, 11 May 1998)	Indian technology commercialisation	Given on 11 May each year	Check tdb.gov.in

PART 7 — PEOPLE AND INSTITUTIONS

About this Part. Inventors, engineers, companies and institutions. Many inventions had several fathers and mothers; we name the main ones and note disputes.

TEC-18 Inventors and engineers

☐ TEC-18.1 India

☐ **TEC-18.1a Aryabhata and Brahmagupta**: mathematics that underlies computing.

☐ **TEC-18.1b Jagadish Chandra Bose (1858 to 1937)**: millimetre-wave radio experiments (1895), crescograph; refused to patent his early work.

☐ **TEC-18.1c M. Visvesvaraya (1861 to 1962)**: engineer of the Krishna Raja Sagara dam and automatic flood gates; Bharat Ratna 1955; his birthday, 15 September, is Engineers' Day.

☐ **TEC-18.1d Homi Bhabha and Vikram Sarabhai**: founders of India's nuclear and space programmes.

☐ **TEC-18.1e Satish Dhawan, U. R. Rao and K. Sivan**: ISRO leaders; the Sriharikota launch centre is named after Dhawan.

☐ **TEC-18.1f A. P. J. Abdul Kalam (1931 to 2015)**: SLV-3 project director, missile programme leader, President of India; Bharat Ratna 1997.

☐ **TEC-18.1g Narinder Singh Kapany (1926 to 2020)**: "father of fibre optics"; Padma Vibhushan 2021 (posthumous).

☐ **TEC-18.1h Sam Pitroda, Nandan Nilekani and the India Stack team**: C-DOT telephone exchanges; Aadhaar and digital public infrastructure.

☐ **TEC-18.1i Ajay Bhatt and Vinod Dham**: co-inventor of USB at Intel; a key engineer of the Pentium chip.

☐ **TEC-18.1j Women in Indian technology**: Tessy Thomas ("Missile Woman", Agni), Ritu Karidhal and M. Vanitha (Chandrayaan and Mangalyaan), Roddam Narasimha's colleagues, Kalpana Chawla (astronaut), Sudha Murty (Telco's first woman engineer).

☐ **TEC-18.1k Grassroots innovators**: the National Innovation Foundation (2000) and Anil Gupta's Honey Bee Network; Mansukh Prajapati's Mitticool clay fridge; Arunachalam Muruganantham's low-cost sanitary-pad machine; Saidullah's amphibious bicycle. *Jugaad* (frugal improvisation) is celebrated, with a note that it should not replace safety.

☐ TEC-18.2 The world

☐ **TEC-18.2a Archimedes, Hero of Alexandria, Zhang Heng, Al-Jazari**: ancient and medieval engineers; Al-Jazari's *Book of Knowledge of Ingenious Mechanical Devices* (1206).

☐ **TEC-18.2b Leonardo da Vinci**: notebooks of machines, most never built.

☐ **TEC-18.2c James Watt, George Stephenson, Isambard Kingdom Brunel**: steam, railways, ships.

☐ **TEC-18.2d Michael Faraday, Nikola Tesla, Thomas Edison, Alexander Graham Bell, Guglielmo Marconi**: the electrical age.

☐ **TEC-18.2e The Wright brothers, Henry Ford, Karl Benz, Bertha Benz** (who made the first long car journey, 1888).

☐ **TEC-18.2f Ada Lovelace, Alan Turing, Grace Hopper, John von Neumann**: computing.

☐ **TEC-18.2g Hedy Lamarr**: co-inventor of frequency hopping (1942), an idea behind later wireless systems.

☐ **TEC-18.2h Vint Cerf, Bob Kahn, Tim Berners-Lee**: the internet and the Web.

☐ **TEC-18.2i Steve Jobs, Steve Wozniak, Bill Gates, Linus Torvalds**: personal computing and open source.

☐ **TEC-18.2j Geoffrey Hinton, Yann LeCun, Yoshua Bengio, Fei-Fei Li, Demis Hassabis**: modern AI.

TEC-19 Companies and institutions

☐ **TEC-19.1 Indian institutions**: ISRO, DRDO, BARC, CSIR labs, C-DOT, C-DAC, NPCI (UPI, RuPay), BEL, HAL, BHEL, IITs (IIT Kharagpur, 1951), IISc (1909, founded with J. N. Tata's support), Department of Science and Technology, MeitY.

☐ **TEC-19.2 Indian companies**: Tata group, Infosys, TCS, Wipro, HCL, Mahindra, Bajaj, Reliance Jio, Bharti Airtel, Biocon, Serum Institute, Bharat Biotech, startups in fintech, space (Skyroot, Agnikul) and AI.

☐ **TEC-19.3 Global companies and labs**: Bell Labs, IBM, Intel, TSMC, Samsung, Apple, Microsoft, Google, NVIDIA, ASML, Siemens, Toyota; public labs such as CERN, NASA, ESA, and national standards labs.

☐ **TEC-19.4 Standards bodies**: ISO, IEC, IEEE, ITU (1865, the oldest UN agency), Bureau of Indian Standards (BIS), the W3C.

☐ **TEC-19.5 Museums**: Science Museum (London), Smithsonian (Washington), Deutsches Museum (Munich), Visvesvaraya Industrial and Technological Museum (Bengaluru), National Science Centre (Delhi), Birla Industrial and Technological Museum (Kolkata), the Nehru Science Centre (Mumbai).

PART 8 — IDEAS: PHILOSOPHY, PSYCHOLOGY AND SOCIETY

About this Part. The philosophy and psychology angles of the brief. Ideas are presented fairly, with sources.

TEC-20 Philosophy of technology

☐ **TEC-20.1 Tools as extensions of the body**: Ernst Kapp (1877) and Marshall McLuhan's *Understanding Media* (1964): "the medium is the message"; the wheel extends the foot, the book the eye.

☐ **TEC-20.2 Gandhi on machines**: *Hind Swaraj* (1909) criticised machinery that displaced people; later Gandhi welcomed machines that served people, such as the Singer sewing machine; the *charkha* as a symbol of self-reliance.

☐ **TEC-20.3 Appropriate technology**: E. F. Schumacher, *Small Is Beautiful* (1973), drew on Gandhian ideas; intermediate technology for villages.

☐ **TEC-20.4 Is technology neutral?**: Langdon Winner's "Do Artifacts Have Politics?" (1980); Heidegger's "The Question Concerning Technology" (1954); Lewis Mumford.

☐ **TEC-20.5 Technological determinism versus social shaping**: does technology drive society, or society shape technology? Most historians now say both.

TEC-21 Technology and society

☐ **TEC-21.1 Technology and jobs**: the Luddites (1811 to 1816), automation, AI and work; studies differ on how many jobs will change; new skills and social protection.

☐ **TEC-21.2 Privacy and surveillance**: data protection, India's Supreme Court ruling that privacy is a fundamental right (*Puttaswamy*, 2017), the DPDP Act (2023).

☐ **TEC-21.3 E-waste and the environment**: the Global E-waste Monitor (2024) counted about 62 million tonnes of e-waste in 2022, with under a quarter documented as recycled; mining for critical minerals; energy use of data centres.

☐ **TEC-21.4 Repair culture and right to repair**: repair cafés; the EU's right-to-repair rules; India's Right to Repair portal (2022) under the Department of Consumer Affairs.

☐ **TEC-21.5 Digital well-being**: screen time, attention and sleep; WHO guidance for young children's screen time (2019); notification hygiene.

☐ **TEC-21.6 The digital divide and inclusion**: access by gender, region and income; Indian-language interfaces; accessibility.

☐ **TEC-21.7 AI ethics and safety**: bias, misinformation and deepfakes, copyright, safety research; India's IT Rules amendments on synthetic media; the EU AI Act (2024).

☐ **TEC-21.8 Psychology of technology**: the "uncanny valley" (Masahiro Mori, 1970), the endowment effect with gadgets, variable rewards in apps, technostress.

PART 9 — THE PRACTICAL GUIDE

TEC-22 Choosing, caring for and recycling devices

☐ **TEC-22.1 How to choose a device**: decide your needs first; compare BEE star labels, warranty, service network, repairability and software-update years; read independent reviews with test data.

☐ **TEC-22.2 Care**: keep vents clear, avoid heat, charge with certified chargers, update software, clean dust; descale geysers and kettles.

☐ **TEC-22.3 Electrical safety**: earthing, MCBs and RCCBs (residual-current breakers save lives), ISI-marked wiring, never touch switches with wet hands, unplug during storms.

☐ **TEC-22.4 Battery safety**: do not puncture or overheat lithium batteries; use proper chargers; dispose of them at collection points.

☐ **TEC-22.5 Cyber safety**: strong passwords and a password manager, two-factor authentication, never share OTPs, report fraud at cybercrime.gov.in or helpline 1930.

☐ **TEC-22.6 Right to repair and recycling**: use authorised e-waste collectors (CPCB list); donate working devices; remove personal data before giving away.

☐ **TEC-22.7 Learning technology**: maker spaces, Atal Tinkering Labs in schools (2016), online courses, electronics kits, coding clubs.

☐ **TEC-22.8 Careers**: engineering branches, ITI trades, diploma courses, research, design, repair entrepreneurship.

PART 10 — REVIEWS AND VOICES

TEC-23 Critics' Corner: technology writers and reviewers

☐ **TEC-23.1 Writers and thinkers**: Lewis Mumford (*Technics and Civilization*, 1934), Jared Diamond (*Guns, Germs, and Steel*, 1997), Walter Isaacson (*The Innovators*, 2014), Yuval Noah Harari, Kate Crawford (*Atlas of AI*, 2021), Shoshana Zuboff (*The Age of Surveillance Capitalism*, 2019), Nandan Nilekani (*Imagining India*, 2008).

☐ **TEC-23.2 Reviewers and media**: *Wired*, *The Verge*, *Ars Technica*, *IEEE Spectrum*, *MIT Technology Review*, *Digit* and *Gadgets 360* in India, independent YouTube reviewers who publish test methods.

TEC-24 The critics said, I say (page template)

1. **The device, invention or book in two lines**: what, who, when, price (dated).
2. **What the critics said**: 3 to 5 named views, summarised in my own words, with links and dates.
3. **Where they agree and disagree**.
4. **What I say**: my own view (link to TEC-25).
5. **If you liked this, try next**: three linked items.

TEC-25 My Review Desk (Kritikesh's device review template)

☐ **Header**: product, maker, model, price and date bought or tested.

☐ **My one-line verdict**.

☐ **What it does and how well**: tests I ran, with numbers.

☐ **Build, repairability and software support**.

☐ **Energy use and BEE rating** (for appliances).

☐ **Privacy and safety**: what data it collects.

☐ **Value for money in India**.

☐ **My rating** on a clear five-point scale, and **tags and try-next links**.

PART 11 — NAVIGATION: TAGS, LINKS AND THE SITE MAP

TEC-26 Tag dictionary

☐ **Trunk**: #devices #appliances #mechanical #tools #fields.

☐ **Device**: #smartphone #laptop #tv #router #battery #camera #drone #fridge #ac #fan #mixer.

☐ **Mechanism**: #lever #wheel #pulley #screw #gear #engine #motor #turbine.

☐ **Field**: #textiles #printing #construction #transport #energy #agriculture #food #water #medicine #communication #computing #ai #space #defence #materials #biotech #environment #fintech #edtech #assistive.

☐ **Era**: #prehistoric #ancient #medieval #industrial #electrical #electronic #digital #ai.

☐ **Region**: #india and each state, and each country.

☐ **Theme**: #inventors #awards #philosophy #privacy #ewaste #repair #safety #jobs #ethics.

☐ **Level**: #beginner #enthusiast #engineer.

TEC-27 Cross-links and the website site map

☐ **TEC-27.1 Cross-topic links**: Music Instruments (electronic instruments: INS); Movies (film technology: MOV); Books (printing: BOK); Sports (sports technology and esports: SPT); Gardening (irrigation and agri-tech: GRD); Cooking (kitchen appliances, food technology: COK); the craft indexes (looms, sewing and knitting machines: CRO, KNT, EMB, SEW); the Curriculum Index (Physics, Chemistry, Computer Science, History of Science) and the Exam Guide (RPSC/RSSB science and technology current affairs, ISRO and defence questions).

☐ **TEC-27.2 Suggested web addresses**: /technology/ (home, Gateway), /technology/stone-to-smartphone/<leaf>/ , /technology/devices/<group>/<device>/ , /technology/appliances/<appliance>/ , /technology/mechanical/<machine>/ , /technology/tools/<tool>/ , /technology/fields/<field>/<topic>/ , /technology/timeline/<era>/ , /technology/people/<name>/ , /technology/institutions/<name>/ , /technology/awards/<award>/<year>/ , /technology/ideas/<topic>/ , /technology/guide/<topic>/ , /technology/reviews/<item>/ .

☐ **TEC-27.3 Search phrases to own**: "how does a fridge work", "how the internet works simple", "six simple machines examples", "history of technology timeline", "Indian inventions list", "ISRO missions list", "Turing Award 2025 winner", "Nobel Prize 2026 physics", "how to dispose of e-waste in India", with Hindi lines ("fridge kaise kaam karta hai", "internet kaise chalta hai", "bharat ke avishkar").

PART 12 — SOURCES, CREDITS AND EDITOR'S NOTES

TEC-S Sources and credits

Knowledge belongs to the people who made it. These are the sources this index stands on; every page on the website will cite its own.

☐ **Reference and museums**: Encyclopaedia Britannica; Smithsonian Institution; Science Museum, London; IEEE History Center and Milestones; national patent offices; US National Academy of Engineering, *Greatest Engineering Achievements of the 20th Century* (2000).

☐ **India**: ISRO; DRDO; MeitY; Department of Telecommunications and C-DOT; NPCI; BEE; CPCB; CSIR-NPL; National Innovation Foundation; PIB releases.

☐ **History and ideas**: Lewis Mumford (1934); McLuhan (1964); Schumacher (1973); Winner (1980); Gandhi, *Hind Swaraj* (1909); Isaacson (2014); Harmand and colleagues on Lomekwi tools (*Nature*, 2015); Pappu and colleagues on Attirampakkam (*Science*, 2011); the Global E-waste Monitor (2024).

☐ **Latest results checked (13 November 2026)**: Nobel prizes 2026 (nobelprize.org and press); ACM Turing Award 2025 (announced 18 March 2026); IEEE Medal of Honor 2026; QEPrize 2026; Millennium Technology Prize (2024 winner; 2027 nominations open); Rashtriya Vigyan Puraskar 2025 (PIB, October 2025).

TEC-E Editor's notes (corrections and decisions on the brief)

1. **Two trunks**: the brief's "devices" and "technology by field" are kept as two linked trunks, so each item has one home and many links.
2. **Disputed inventions** (telephone, radio, light bulb, television) name several contributors.
3. **Jugaad** is celebrated with a safety note.
4. **Defence technology** is treated as history and engineering, not weapons advice.
5. **Medical technology** carries a "not medical advice" note.
6. **The Shanti Swarup Bhatnagar Prize** now lives on as the Vigyan Yuva award within the Rashtriya Vigyan Puraskar (from 2024); the brief's name is kept as a search term.
7. **State of the art** is dated November 2026; device models are not listed because they change monthly.
8. **The 7-field register** is split into two linked tables to fit the page.
9. **Not stated because not verified**: the latest Draper Prize; it will be added on the website.

TEC-T My Technology Tracker

Use the ☐ boxes across the booklet to tick what you have used, opened up, repaired or explored. A simple tracker for each Part: Part · Sections · Explored · Date started · Date finished. Keep a maker's log: date, device, what I learned, what I fixed.

Progress tracker

Part	Sections	Explored ☐	Started	Finished
PART 0 — THE INTRODUCTION TWIG: HOW DID A SHARPENED STONE BECOME A SMARTPHONE?	9			
PART 1 — THE MASTER MAP OF TECHNOLOGY	1			
PART 2 — TRUNK A: ELECTRONIC DEVICES AND HOME APPLIANCES	2			

PART 3 — TRUNK A: MECHANICAL DEVICES, MACHINES, VEHICLES AND TOOLS	3			
PART 4 — TRUNK B: TECHNOLOGY BY FIELD (1)	4			
PART 5 — TRUNK B: TECHNOLOGY BY FIELD (2)	4			
PART 6 — REGISTERS	3			
PART 7 — PEOPLE AND INSTITUTIONS	2			
PART 8 — IDEAS: PHILOSOPHY, PSYCHOLOGY AND SOCIETY	2			
PART 9 — THE PRACTICAL GUIDE	1			
PART 10 — REVIEWS AND VOICES	3			
PART 11 — NAVIGATION: TAGS, LINKS AND THE SITE MAP	2			
PART 12 — SOURCES, CREDITS AND EDITOR'S NOTES	3			